Department of Mathematics and Natural Sciences

Final Examination
Semester: Summer 2025
Course Title: MATHEMATICS I: Differential Calculus & Coordinate Geometry
Course Code: MAT-110
Time: 1 hour 20 minutes **Date: September 15, 2025**
Total Marks: 35

SOLUTION & MARKING SCHEME

Created by: SAKAL ROY [SKY]

✅ NOTE:

TO APPRECIATE YOUR MATHEMATICAL THINKING, ANY VALID MATHEMATICAL WAY OF SOLVING WILL BE TREATED EQUALLY AND ASSESSED ACCORDINGLY, AS LONG AS IT'S CONSISTENT WITH THE PROVIDED INSTRUCTIONS OF THE QUESTION.

IN THIS ASSESSMENT PROCESS EVERY PROBLEM WILL BE ASSESSED BASED ON 3 CRITERIA:
A. MATHEMATICAL THINKING
B. MATHEMATICAL KNOWLEDGE
C. MATHEMATICAL SKILL

✅ PERCENTAGE SCHEME

- **N/A = Not Assessed / Not Answered**
- **0% = Incorrect**
- **50% = Partially Correct**
- **80% = Mostly Correct**
- **100% = Perfect**

✅ INDEX

QUESTION_1:

Q1. Let $f(u,v) = 5\cos u - 3\sin(uv)$, $u(x,y,z) = 2x + y + z$, $v(x,y,z) = x^2 - zy^2$. [7]
Use appropriate forms of chain rule to find $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$.

Solution:

We have

$$f(u,v) = 5\cos u - 3\sin(uv), \qquad u(x,y,z) = 2x + y + z, \qquad v(x,y,z) = x^2 - zy^2.$$

Chain rule for functions of several variables

$$\boxed{\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x}, \qquad \frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y}, \qquad \frac{\partial f}{\partial z} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial z} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial z}.}$$

Step 1 — partials of f (w.r.t. u, v)

$$\frac{\partial f}{\partial u} = -5\sin u \; - \; 3v\cos(uv), \qquad \frac{\partial f}{\partial v} = -3u\cos(uv).$$

Step 2 — partials of u and new v

$$\frac{\partial u}{\partial x} = 2, \quad \frac{\partial u}{\partial y} = 1, \quad \frac{\partial u}{\partial z} = 1,$$

$$\frac{\partial v}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = -2zy, \quad \frac{\partial v}{\partial z} = -y^2.$$

Step 3 — apply the chain rule

$\dfrac{\partial f}{\partial x}$

$$\frac{\partial f}{\partial x} = \big(-5\sin u - 3v\cos(uv)\big)\cdot 2 \; + \; \big(-3u\cos(uv)\big)\cdot(2x),$$

$$\boxed{\frac{\partial f}{\partial x} = -10\sin u \; - \; 6v\cos(uv) \; - \; 6xu\cos(uv).}$$

$\dfrac{\partial f}{\partial y}$

$$\frac{\partial f}{\partial y} = \big(-5\sin u - 3v\cos(uv)\big)\cdot 1 \;+\; \big(-3u\cos(uv)\big)\cdot(-2zy),$$

$$\boxed{\frac{\partial f}{\partial y} = -5\sin u \;-\; 3v\cos(uv) \;+\; 6uyz\cos(uv).}$$

$\dfrac{\partial f}{\partial z}$

$$\frac{\partial f}{\partial z} = \big(-5\sin u - 3v\cos(uv)\big)\cdot 1 \;+\; \big(-3u\cos(uv)\big)\cdot(-y^2),$$

$$\boxed{\frac{\partial f}{\partial z} = -5\sin u \;-\; 3v\cos(uv) \;+\; 3uy^2\cos(uv).}$$

QUESTION_2:

Q2. Find the 1st degree ($L(x, y)$) and 2nd degree ($Q(x, y)$) Taylor polynomial approximations respectively of the function, $f(x, y) = \sqrt{2x - y}$, near the point $(1, -2)$. [7]

Solution:

We want the **first-degree Taylor polynomial** $L(x, y)$ and the **second-degree Taylor polynomial** $Q(x, y)$ of

$$f(x, y) = \sqrt{2x - y}, \quad \text{about } (a, b) = (1, -2).$$

Step 1. Evaluate the function at $(1, -2)$

$$f(1, -2) = \sqrt{2(1) - (-2)} = \sqrt{4} = 2.$$

1. Compute the value $2x - y$ at $(1, -2)$

$$2x - y\Big|_{(1,-2)} = 2(1) - (-2) = 2 + 2 = 4.$$

So everywhere below you can replace $(2x - y)$ by 4 when evaluating at the point.

Step 2. First-order partial derivatives

$$f_x(x, y) = \frac{\partial}{\partial x}\left((2x - y)^{1/2}\right) = \frac{1}{2}(2x - y)^{-1/2} \cdot 2 = (2x - y)^{-1/2}.$$

$$f_y(x, y) = \frac{\partial}{\partial y}\left((2x - y)^{1/2}\right) = \frac{1}{2}(2x - y)^{-1/2} \cdot (-1) = -\tfrac{1}{2}(2x - y)^{-1/2}.$$

Now substitute $x = 1, y = -2$:

$$f_x(1, -2) = (2 \cdot 1 - (-2))^{-1/2} = 4^{-1/2}.$$

Evaluate $4^{-1/2}$ step-by-step:

$$4^{-1/2} = \frac{1}{4^{1/2}} = \frac{1}{\sqrt{4}} = \frac{1}{2}.$$

So

$$\boxed{f_x(1, -2) = \tfrac{1}{2}}$$

Substitute $x = 1, y = -2$:

$$f_y(1, -2) = -\tfrac{1}{2} \cdot 4^{-1/2}.$$

We already found $4^{-1/2} = \frac{1}{2}$, so

$$f_y(1, -2) = -\tfrac{1}{2} \cdot \tfrac{1}{2} = -\tfrac{1}{4}.$$

So

$$\boxed{f_y(1, -2) = -\tfrac{1}{4}}$$

Step 3. Second-order partial derivatives

$$f_{xx}(x, y) = \frac{\partial}{\partial x}\left((2x - y)^{-1/2}\right) = -\tfrac{1}{2}(2x - y)^{-3/2} \cdot 2 = -(2x - y)^{-3/2}.$$

$$f_{xy}(x, y) = \frac{\partial}{\partial y}\left((2x - y)^{-1/2}\right) = -\tfrac{1}{2}(2x - y)^{-3/2} \cdot (-1) = \tfrac{1}{2}(2x - y)^{-3/2}.$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y}\Big(-\tfrac{1}{2}(2x - y)^{-1/2}\Big) = -\tfrac{1}{2} \cdot \Big(-\tfrac{1}{2}\Big)(2x - y)^{-3/2} \cdot (-1) = -\tfrac{1}{4}(2x - y)^{-3/2}.$$

We will need $(2x-y)^{-3/2}$ at the point; compute it first:

$$(2x-y)^{-3/2}\Big|_{(1,-2)} = 4^{-3/2}.$$

Evaluate $4^{-3/2}$ step-by-step:

$$4^{3/2} = \left(4^{1/2}\right)^3 = (\sqrt{4})^3 = 2^3 = 8, \qquad \text{so} \qquad 4^{-3/2} = \frac{1}{8}.$$

Substitute the point:

$$f_{xx}(1,-2) = -4^{-3/2} = -\frac{1}{8}.$$

So

$$\boxed{f_{xx}(1,-2) = -\tfrac{1}{8}}$$

Substitute the point:

$$f_{xy}(1,-2) = \tfrac{1}{2}\cdot 4^{-3/2} = \tfrac{1}{2}\cdot\tfrac{1}{8} = \tfrac{1}{16}.$$

So

$$\boxed{f_{xy}(1,-2) = \tfrac{1}{16}}$$

(As a check, $f_{yx}(1,-2) = f_{xy}(1,-2) = 1/16$ by equality of mixed partials under regularity conditions.)

Substitute the point:

$$f_{yy}(1,-2) = -\tfrac{1}{4}\cdot 4^{-3/2} = -\tfrac{1}{4}\cdot\tfrac{1}{8} = -\tfrac{1}{32}.$$

So

$$\boxed{f_{yy}(1,-2) = -\tfrac{1}{32}}$$

Step 4. First-degree Taylor polynomial

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

$$L(x, y) = 2 + \tfrac{1}{2}(x - 1) - \tfrac{1}{4}(y + 2).$$

Simplify:

$$L(x, y) = 1 + \tfrac{1}{2}x - \tfrac{1}{4}y.$$

Step 5. Second-degree Taylor polynomial

$$Q(x, y) = L(x, y) + \tfrac{1}{2}\Big[f_{xx}(a, b)(x - a)^2 + 2f_{xy}(a, b)(x - a)(y - b) + f_{yy}(a, b)(y - b)^2\Big].$$

Substitute values:

$$Q(x, y) = 2 + \tfrac{1}{2}(x - 1) - \tfrac{1}{4}(y + 2) + \tfrac{1}{2}\Big[- \tfrac{1}{8}(x - 1)^2 + 2 \cdot \tfrac{1}{16}(x - 1)(y + 2) - \tfrac{1}{32}(y + 2)^2\Big].$$

Simplify coefficients:

$$Q(x, y) = 2 + \tfrac{1}{2}(x - 1) - \tfrac{1}{4}(y + 2) - \tfrac{1}{16}(x - 1)^2 + \tfrac{1}{16}(x - 1)(y + 2) - \tfrac{1}{64}(y + 2)^2.$$

✅ Final Results:

$$L(x, y) = 1 + \tfrac{1}{2}x - \tfrac{1}{4}y,$$

$$Q(x, y) = 2 + \tfrac{1}{2}(x - 1) - \tfrac{1}{4}(y + 2) - \tfrac{1}{16}(x - 1)^2 + \tfrac{1}{16}(x - 1)(y + 2) - \tfrac{1}{64}(y + 2)^2.$$

QUESTION_3:

Q3. Find all relative extrema and saddle points (if any) for the function: [7]

$$f(x, y) = x^4 + y^4 - 4xy + 1.$$

Solution:

See here:

https://www.emathhelp.net/calculators/calculus-3/critical-points-extrema-saddle-points-calculator/?f=x%5E4%2By%5E4-4xy%2B1

The first step is to find all the first-order partial derivatives:

$\frac{\partial}{\partial x}\left(x^4 - 4xy + y^4 + 1\right) = 4x^3 - 4y$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial y}\left(x^4 - 4xy + y^4 + 1\right) = -4x + 4y^3$ (for steps, see partial derivative calculator).

Next, solve the system $\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases}$, or $\begin{cases} 4x^3 - 4y = 0 \\ -4x + 4y^3 = 0 \end{cases}$.

$$\text{or} \quad \begin{aligned} y &= x^3 \\ x &= y^3 \end{aligned} \qquad (2)$$

Substituting the top equation in the bottom yields $x = (x^3)^3$ or, equivalently, $x^9 - x = 0$ or $x(x^8 - 1) = 0$, which has solutions $x = 0$, $x = 1$, $x = -1$. Substituting these values in the top equation of (2), we obtain the corresponding y-values $y = 0$, $y = 1$, $y = -1$. Thus, the critical points of f are $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

The system has the following real solutions: $(x, y) = (-1, -1)$, $(x, y) = (0, 0)$, $(x, y) = (1, 1)$.

Now, let's try to classify them.

Find all the second-order partial derivatives:

$\frac{\partial^2}{\partial x^2}\left(x^4 - 4xy + y^4 + 1\right) = 12x^2$ (for steps, see partial derivative calculator).

$\frac{\partial^2}{\partial y \partial x}\left(x^4 - 4xy + y^4 + 1\right) = -4$ (for steps, see partial derivative calculator).

$\frac{\partial^2}{\partial y^2}\left(x^4 - 4xy + y^4 + 1\right) = 12y^2$ (for steps, see partial derivative calculator).

Define the expression $D = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial y \partial x}\right)^2 = 144x^2y^2 - 16.$

Since $D(-1, -1) = 128$ is greater than 0 and $\frac{\partial^2}{\partial x^2}\left(x^4 - 4xy + y^4 + 1\right)|_{((x,y)=(-1,-1))} = 12$ is greater than 0, it can be stated that $(-1, -1)$ is a relative minimum.

Since $D(0, 0) = -16$ is less than 0, it can be stated that $(0, 0)$ is a saddle point.

Since $D(1, 1) = 128$ is greater than 0 and $\frac{\partial^2}{\partial x^2}\left(x^4 - 4xy + y^4 + 1\right)|_{((x,y)=(1,1))} = 12$ is greater than 0, it can be stated that $(1, 1)$ is a relative minimum.

ANSWER

Relative Maxima

No relative maxima.

Relative Minima

$(x, y) = (-1, -1)$ A , $f(-1, -1) = -1$ A

$(x, y) = (1, 1)$ A , $f(1, 1) = -1$ A

Saddle Points

$(x, y) = (0, 0)$ A , $f(0, 0) = 1$ A

QUESTION_4:

Q4. Use Lagrange multiplier to find maximum and minimum values of the function, $f(x,y,z) = xyz$ subject to the constraint $x^3 + y^3 + z^3 = 27$. [7]

Solution:

See here:

https://www.emathhelp.net/calculators/calculus-3/lagrange-multipliers-calculator/?f=xyz&g=x%5E3+%2By%5E3+%2Bz%5E3+%3D27&h=

Rewrite the constraint $x^3 + y^3 + z^3 = 27$ as $x^3 + y^3 + z^3 - 27 = 0$.

Form the Lagrangian: $L(x, y, z, \lambda) = xyz + \lambda\left(x^3 + y^3 + z^3 - 27\right)$.

Find all the first-order partial derivatives:

$\frac{\partial}{\partial x}\left(xyz + \lambda\left(x^3 + y^3 + z^3 - 27\right)\right) = 3\lambda x^2 + yz$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial y}\left(xyz + \lambda\left(x^3 + y^3 + z^3 - 27\right)\right) = 3\lambda y^2 + xz$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial z}\left(xyz + \lambda\left(x^3 + y^3 + z^3 - 27\right)\right) = 3\lambda z^2 + xy$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial \lambda}\left(xyz + \lambda\left(x^3 + y^3 + z^3 - 27\right)\right) = x^3 + y^3 + z^3 - 27$ (for steps, see partial derivative calculator).

Next, solve the system $\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ \frac{\partial L}{\partial \lambda} = 0 \end{cases}$, or $\begin{cases} 3\lambda x^2 + yz = 0 \\ 3\lambda y^2 + xz = 0 \\ 3\lambda z^2 + xy = 0 \\ x^3 + y^3 + z^3 - 27 = 0 \end{cases}$.

Step 1: Analyze the first three equations

The first three equations are symmetric in x, y, z:

$$3\lambda x^2 + yz = 0 \quad (1)$$
$$3\lambda y^2 + xz = 0 \quad (2)$$
$$3\lambda z^2 + xy = 0 \quad (3)$$

Step 2: Try the case $x = y = z$

Let $x = y = z = t$. Substitute into the first equation:

$$3\lambda t^2 + t \cdot t = 0 \implies 3\lambda t^2 + t^2 = 0 \implies t^2(3\lambda + 1) = 0$$

So either $t = 0$ or $3\lambda + 1 = 0 \implies \lambda = -\frac{1}{3}$.

Step 3: Use the fourth equation with $x = y = z = t$

$$x^3 + y^3 + z^3 - 27 = 3t^3 - 27 = 0 \implies 3t^3 = 27 \implies t^3 = 9 \implies t = \sqrt[3]{9}$$

Step 4: Check consistency with λ

From Step 2, if $t \neq 0$, then $\lambda = -\frac{1}{3}$.

Step 5: Write the solution

$$x = y = z = \sqrt[3]{9}, \quad \lambda = -\frac{1}{3}$$

Step 6: Check if $t = 0$ is a solution

If $t = 0$, then from the fourth equation:

$$0 + 0 + 0 - 27 = -27 \neq 0$$

So $t = 0$ is not a solution.

Final answer:

$$\boxed{x = y = z = \sqrt[3]{9}, \quad \lambda = -\frac{1}{3}}$$

Let's explore the possibility of asymmetric solutions (x, y, z) where x, y, z are not all equal.

Step 1: Recall the system

$$\begin{cases} 3\lambda x^2 + yz = 0 \\ 3\lambda y^2 + xz = 0 \\ 3\lambda z^2 + xy = 0 \\ x^3 + y^3 + z^3 = 27 \end{cases}$$

Step 2: Express λ from each of the first three equations

From the first equation:

$$3\lambda x^2 = -yz \implies \lambda = -\frac{yz}{3x^2}$$

Similarly, from the second:

$$\lambda = -\frac{xz}{3y^2}$$

From the third:

$$\lambda = -\frac{xy}{3z^2}$$

Since λ is the same in all three, equate these:

$$-\frac{yz}{3x^2} = -\frac{xz}{3y^2} \implies \frac{yz}{x^2} = \frac{xz}{y^2}$$

Multiply both sides by x^2y^2:

$$yzy^2 = xzx^2 \implies y^3z = x^3z$$

If $z \neq 0$, divide both sides by z:

$$y^3 = x^3 \implies y = x \quad \text{(assuming real numbers)}$$

Similarly, equate the first and third expressions for λ:

$$-\frac{yz}{3x^2} = -\frac{xy}{3z^2} \implies \frac{yz}{x^2} = \frac{xy}{z^2}$$

Multiply both sides by x^2z^2:

$$yzz^2 = xyx^2 \implies yz^3 = x^3y$$

If $y \neq 0$, divide both sides by y:

$$z^3 = x^3 \implies z = x$$

Step 3: Conclusion from Step 2

From the above, if none of x, y, z are zero, then $x = y = z$.

Step 4: Consider the case where some variables are zero

Suppose one of x, y, z is zero, say $z = 0$.
From the first equation:

$$3\lambda x^2 + y \cdot 0 = 0 \implies 3\lambda x^2 = 0$$

If $x \neq 0$, then $\lambda = 0$.
From the second equation:

$$3\lambda y^2 + x \cdot 0 = 0 \implies 3\lambda y^2 = 0$$

If $y \neq 0$, then $\lambda = 0$.
From the third equation:

$$3\lambda 0^2 + xy = 0 \implies xy = 0$$

So either $x = 0$ or $y = 0$.

Step 5: Check the fourth equation with $z = 0$

$$x^3 + y^3 + 0 - 27 = 0 \implies x^3 + y^3 = 27$$

Step 6: Possible asymmetric solutions with zeros

- If $z = 0$, and $x = 0$, then $y^3 = 27 \implies y = 3$.
- If $z = 0$, and $y = 0$, then $x^3 = 27 \implies x = 3$.

So possible solutions are:

$$(0, 3, 0), \quad (3, 0, 0), \quad (0, 0, 3)$$

Step 7: Check if these satisfy the first three equations with $\lambda = 0$

For $(3, 0, 0)$:

- First equation: $3 \cdot 0 \cdot 3^2 + 0 \cdot 0 = 0$ ✓
- Second equation: $3 \cdot 0 \cdot 0^2 + 3 \cdot 0 = 0$ ✓
- Third equation: $3 \cdot 0 \cdot 0^2 + 3 \cdot 0 = 0$ ✓

Similarly for the others.

Summary:

- If none of x, y, z are zero, then $x = y = z = \sqrt[3]{9}$ and $\lambda = -\frac{1}{3}$.
- If some variables are zero, asymmetric solutions exist such as $(3, 0, 0)$, $(0, 3, 0)$, $(0, 0, 3)$ with $\lambda = 0$.

The system has the following real solutions: $(x, y, z) = (0, 0, 3)$, $(x, y, z) = (0, 3, 0)$, $(x, y, z) = (3, 0, 0)$, $(x, y, z) = \left(3^{\frac{2}{3}}, 3^{\frac{2}{3}}, 3^{\frac{2}{3}}\right)$.

$$f(0, 0, 3) = 0$$

$$f(0, 3, 0) = 0$$

$$f(3, 0, 0) = 0$$

$$f\left(3^{\frac{2}{3}}, 3^{\frac{2}{3}}, 3^{\frac{2}{3}}\right) = 9$$

Thus, the minimum value is 0, and the maximum value is 9.

ANSWER

Maximum

9 A **at** $(x, y, z) = \left(3^{\frac{2}{3}}, 3^{\frac{2}{3}}, 3^{\frac{2}{3}}\right) \approx (2.080083823051904, 2.080083823051904, 2.080083823051904)$ A .

Minimum

0 A **at** $(x, y, z) = (0, 0, 3), (x, y, z) = (0, 3, 0), (x, y, z) = (3, 0, 0)$ A .

QUESTION_5:

Q5. Transform the conic $9x^2 - 54x + y^2 + 2y + 46 = 0$ into its standard form. Then find its centre, eccentricity, foci, vertices, equations of directrices and length of latus rectum. [7]

Solution:

See here:

https://www.emathhelp.net/calculators/algebra-2/ellipse-calculator/?type=f&f=9x%5E2+-54x+%2By%5E2+%2B2y%2B46%3D0&cx=&cy=&f1x=&f1y=&f2x=&f2y=&v1x=&v1y=&v2x=&v2y=&cv1x=&cv1y=&cv2x=&cv2y=&ma=&mi=&e=&a=&d1=&d2=&p1x=&p1y=&p2x=&p2y=&p3x=&p3y=&p4x=&p4y=

The equation of an ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, where (h, k) is the center, b and a are the lengths of the semi-major and the semi-minor axes.

Our ellipse in this form is $\frac{(x-3)^2}{4} + \frac{(y-(-1))^2}{36} = 1$.

Thus, $h = 3$, $k = -1$, $a = 2$, $b = 6$.

The standard form is $\frac{(x-3)^2}{2^2} + \frac{(y+1)^2}{6^2} = 1$.

The vertex form is $\frac{(x-3)^2}{4} + \frac{(y+1)^2}{36} = 1$.

The general form is $9x^2 - 54x + y^2 + 2y + 46 = 0$.

The linear eccentricity (focal distance) is $c = \sqrt{b^2 - a^2} = 4\sqrt{2}$.

The eccentricity is $e = \frac{c}{b} = \frac{2\sqrt{2}}{3}$.

The first focus is $(h, k - c) = (3, -4\sqrt{2} - 1)$.

The second focus is $(h, k + c) = (3, -1 + 4\sqrt{2})$.

The first vertex is $(h, k - b) = (3, -7)$.

The second vertex is $(h, k + b) = (3, 5)$.

The first directrix is $y = k - \frac{b^2}{c} = -\frac{2+9\sqrt{2}}{2}$.

The second directrix is $y = k + \frac{b^2}{c} = \frac{-2+9\sqrt{2}}{2}$.

The latera recta are the lines parallel to the minor axis that pass through the foci.

The first latus rectum is $y = -4\sqrt{2} - 1$.

The second latus rectum is $y = -1 + 4\sqrt{2}$.

The endpoints of the first latus rectum can be found by solving the system $\begin{cases} 9x^2 - 54x + y^2 + 2y + 46 = 0 \\ y = -4\sqrt{2} - 1 \end{cases}$ (for steps, see system of equations calculator).

The endpoints of the first latus rectum are $\left(\frac{7}{3}, -4\sqrt{2} - 1\right)$, $\left(\frac{11}{3}, -4\sqrt{2} - 1\right)$.

The endpoints of the second latus rectum can be found by solving the system $\begin{cases} 9x^2 - 54x + y^2 + 2y + 46 = 0 \\ y = -1 + 4\sqrt{2} \end{cases}$ (for steps, see system of equations calculator).

The endpoints of the second latus rectum are $\left(\frac{7}{3}, -1 + 4\sqrt{2}\right)$, $\left(\frac{11}{3}, -1 + 4\sqrt{2}\right)$.

The length of the latera recta (focal width) is $\frac{2a^2}{b} = \frac{4}{3}$.

QUESTION_6:

Q6. (a) Let $f(x,y,z) = xy^2z^3 + \sin^{-1}(x\sqrt{z})$. Find f_x, f_{xz} and f_{xzy}. [1.5+1.5+1]

(b) Convert from cylindrical to spherical coordinates: $(4, \frac{5\pi}{6}, 4)$. [3]

Solution:

6(a):

See here:
https://www.emathhelp.net/calculators/calculus-3/partial-derivative-calculator/?f=x*y%5E2*z%5E3%2B+arcsin%28x+*+sqrt%28z%29%29&var=x

Your input: find $\frac{\partial}{\partial x}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right)$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial x}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = \left(\frac{\partial}{\partial x}\left(xy^2z^3\right) + \frac{\partial}{\partial x}\left(\operatorname{asin}\left(x\sqrt{z}\right)\right)\right)$$

Write the function $\operatorname{asin}\left(x\sqrt{z}\right)$ as a composition of the two functions $u = g = x\sqrt{z}$ and $f(u) = \operatorname{asin}(u)$.

Apply the chain rule $\frac{\partial}{\partial x}(f(g)) = \frac{\partial}{\partial u}(f(u)) \cdot \frac{\partial}{\partial x}(g)$:

$$\frac{\partial}{\partial x}\left(\operatorname{asin}\left(x\sqrt{z}\right)\right) + \frac{\partial}{\partial x}\left(xy^2z^3\right) = \frac{\partial}{\partial u}\left(\operatorname{asin}(u)\right)\frac{\partial}{\partial x}\left(x\sqrt{z}\right) + \frac{\partial}{\partial x}\left(xy^2z^3\right)$$

The derivative of inverse sine is $\frac{\partial}{\partial u}\left(\operatorname{asin}(u)\right) = \frac{1}{\sqrt{1-u^2}}$:

$$\frac{\partial}{\partial u}\left(\operatorname{asin}(u)\right)\frac{\partial}{\partial x}\left(x\sqrt{z}\right) + \frac{\partial}{\partial x}\left(xy^2z^3\right) = \frac{1}{\sqrt{1-u^2}}\frac{\partial}{\partial x}\left(x\sqrt{z}\right) + \frac{\partial}{\partial x}\left(xy^2z^3\right)$$

Return to the old variable:

$$\frac{1}{\sqrt{1-u^2}}\frac{\partial}{\partial x}\left(x\sqrt{z}\right) + \frac{\partial}{\partial x}\left(xy^2z^3\right) = \frac{1}{\sqrt{1-x\sqrt{z}^2}}\frac{\partial}{\partial x}\left(x\sqrt{z}\right) + \frac{\partial}{\partial x}\left(xy^2z^3\right)$$

Apply the constant multiple rule $\frac{\partial}{\partial x}(c \cdot f) = c \cdot \frac{\partial}{\partial x}(f)$ with $c = \sqrt{z}$ and $f = x$:

$$\frac{\partial}{\partial x}\left(xy^2z^3\right) + \frac{\frac{\partial}{\partial x}\left(x\sqrt{z}\right)}{\sqrt{-x^2z+1}} = \frac{\partial}{\partial x}\left(xy^2z^3\right) + \frac{\sqrt{z}\frac{\partial}{\partial x}(x)}{\sqrt{-x^2z+1}}$$

Apply the power rule $\frac{\partial}{\partial x}(x^n) = n \cdot x^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial x}(x) = 1$:

$$\frac{\sqrt{z}\frac{\partial}{\partial x}(x)}{\sqrt{-x^2z+1}} + \frac{\partial}{\partial x}\left(xy^2z^3\right) = \frac{\sqrt{z}1}{\sqrt{-x^2z+1}} + \frac{\partial}{\partial x}\left(xy^2z^3\right)$$

Apply the constant multiple rule $\frac{\partial}{\partial x}(c \cdot f) = c \cdot \frac{\partial}{\partial x}(f)$ with $c = y^2z^3$ and $f = x$:

$$\frac{\sqrt{z}}{\sqrt{-x^2z+1}} + \frac{\partial}{\partial x}\left(xy^2z^3\right) = \frac{\sqrt{z}}{\sqrt{-x^2z+1}} + y^2z^3\frac{\partial}{\partial x}(x)$$

Apply the power rule $\frac{\partial}{\partial x}(x^n) = n \cdot x^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial x}(x) = 1$:

$$y^2z^3\frac{\partial}{\partial x}(x) + \frac{\sqrt{z}}{\sqrt{-x^2z+1}} = y^2z^3 1 + \frac{\sqrt{z}}{\sqrt{-x^2z+1}}$$

Thus, $\frac{\partial}{\partial x}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = y^2z^3 + \frac{\sqrt{z}}{\sqrt{-x^2z+1}}$

Answer: $\frac{\partial}{\partial x}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = y^2z^3 + \frac{\sqrt{z}}{\sqrt{-x^2z+1}}$

See Here:
https://www.emathhelp.net/calculators/calculus-3/partial-derivative-calculator/?f=x*y%5E2*z%5E3%2B+asin+%28x+*+sqrt+%28z%29%29&var=x%2Cz

$$\frac{\partial}{\partial x}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = y^2z^3 + \frac{\sqrt{z}}{\sqrt{-x^2z+1}}$$

NEXT, $\frac{\partial^2}{\partial x \partial z}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = \frac{\partial}{\partial z}\left(\frac{\partial}{\partial x}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right)\right)$

$$= \frac{\partial}{\partial z}\left(y^2z^3 + \frac{\sqrt{z}}{\sqrt{-x^2z+1}}\right)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial z}\left(y^2z^3 + \frac{\sqrt{z}}{\sqrt{-x^2z+1}}\right) = \left(\frac{\partial}{\partial z}\left(y^2z^3\right) + \frac{\partial}{\partial z}\left(\frac{\sqrt{z}}{\sqrt{-x^2z+1}}\right)\right)$$

Apply the quotient rule $\frac{\partial}{\partial z}\left(\frac{f}{g}\right) = \frac{1}{g^2}\left(\frac{\partial}{\partial z}(f)\cdot g - f\cdot\frac{\partial}{\partial z}(g)\right)$ with $f = \sqrt{z}$ and $g = \sqrt{-x^2z+1}$:

$$\frac{\partial}{\partial z}\left(\frac{\sqrt{z}}{\sqrt{-x^2z+1}}\right) + \frac{\partial}{\partial z}\left(y^2z^3\right) = \left(\frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2z+1}\right) + \frac{\partial}{\partial z}\left(\sqrt{z}\right)\sqrt{-x^2z+1}}{-x^2z+1}\right)$$

$$+ \frac{\partial}{\partial z}\left(y^2z^3\right)$$

Apply the power rule $\frac{\partial}{\partial z}(z^n) = n \cdot z^{-1+n}$ with $n = \frac{1}{2}$:

$$\frac{\partial}{\partial z}(y^2 z^3) + \frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2 z+1}\right) + \sqrt{-x^2 z+1}\frac{\partial}{\partial z}\left(\sqrt{z}\right)}{-x^2 z+1} = \frac{\partial}{\partial z}(y^2 z^3)$$
$$+ \frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2 z+1}\right) + \sqrt{-x^2 z+1}\frac{\partial}{\partial z}\left(z^{\frac{1}{2}}\right)}{-x^2 z+1} = \frac{\partial}{\partial z}(y^2 z^3)$$
$$+ \frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2 z+1}\right) + \sqrt{-x^2 z+1}\left(\frac{z^{-1+\frac{1}{2}}}{2}\right)}{-x^2 z+1} = \frac{\partial}{\partial z}(y^2 z^3)$$
$$+ \frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2 z+1}\right) + \sqrt{-x^2 z+1}\left(\frac{z^{-\frac{1}{2}}}{2}\right)}{-x^2 z+1} = \frac{\partial}{\partial z}(y^2 z^3)$$
$$+ \frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2 z+1}\right) + \frac{\sqrt{-x^2 z+1}}{2\sqrt{z}}}{-x^2 z+1}$$

Write the function $\sqrt{-x^2 z+1}$ as a composition of the two functions $u = g = -x^2 z + 1$ and $f(u) = \sqrt{u}$.

Apply the chain rule $\frac{\partial}{\partial z}(f(g)) = \frac{\partial}{\partial u}(f(u)) \cdot \frac{\partial}{\partial z}(g)$:

$$\frac{\partial}{\partial z}(y^2 z^3) + \frac{-\sqrt{z}\frac{\partial}{\partial z}\left(\sqrt{-x^2 z+1}\right) + \frac{\sqrt{-x^2 z+1}}{2\sqrt{z}}}{-x^2 z+1} = \frac{\partial}{\partial z}(y^2 z^3)$$
$$+ \frac{-\sqrt{z}\frac{\partial}{\partial u}\left(\sqrt{u}\right)\frac{\partial}{\partial z}\left(-x^2 z+1\right) + \frac{\sqrt{-x^2 z+1}}{2\sqrt{z}}}{-x^2 z+1}$$

Apply the power rule $\frac{\partial}{\partial u}(u^n) = n \cdot u^{-1+n}$ with $n = \frac{1}{2}$:

$$\frac{\partial}{\partial z}(y^2z^3) + \frac{-\sqrt{z}\frac{\partial}{\partial u}(\sqrt{u})\frac{\partial}{\partial z}(-x^2z+1) + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}(y^2z^3)$$
$$+ \frac{-\sqrt{z}\frac{\partial}{\partial u}\left(u^{\frac{1}{2}}\right)\frac{\partial}{\partial z}(-x^2z+1) + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}(y^2z^3)$$
$$+ \frac{-\sqrt{z}\left(\frac{u^{-1+\frac{1}{2}}}{2}\right)\frac{\partial}{\partial z}(-x^2z+1) + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}(y^2z^3)$$
$$+ \frac{-\sqrt{z}\left(\frac{u^{-\frac{1}{2}}}{2}\right)\frac{\partial}{\partial z}(-x^2z+1) + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}(y^2z^3) + \frac{\frac{\sqrt{-x^2z+1}}{2\sqrt{z}} - \frac{\sqrt{z}\frac{\partial}{\partial z}(-x^2z+1)}{2\sqrt{u}}}{-x^2z+1}$$

Return to the old variable:

$$\frac{\partial}{\partial z}(y^2z^3) + \frac{-\frac{\sqrt{z}\frac{1}{\sqrt{u}}\frac{\partial}{\partial z}(-x^2z+1)}{2} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}(y^2z^3)$$
$$+ \frac{-\frac{\sqrt{z}\frac{1}{\sqrt{(-x^2z+1)}}\frac{\partial}{\partial z}(-x^2z+1)}{2} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{-\frac{\sqrt{z}\frac{\partial}{\partial z}\left(-x^2z+1\right)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{-\frac{\sqrt{z}\left(\frac{\partial}{\partial z}(1)-\frac{\partial}{\partial z}\left(x^2z\right)\right)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}$$

The derivative of a constant is 0:

$$\frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{-\frac{\sqrt{z}\left(\frac{\partial}{\partial z}(1)-\frac{\partial}{\partial z}\left(x^2z\right)\right)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{-\frac{\sqrt{z}\left((0)-\frac{\partial}{\partial z}\left(x^2z\right)\right)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}$$

Apply the constant multiple rule $\frac{\partial}{\partial z}(c \cdot f) = c \cdot \frac{\partial}{\partial z}(f)$ with $c = x^2$ and $f = z$:

$$\frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{\frac{\sqrt{z}\frac{\partial}{\partial z}\left(x^2z\right)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{\frac{\sqrt{z}x^2\frac{\partial}{\partial z}(z)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}$$

Apply the power rule $\frac{\partial}{\partial z}(z^n) = n \cdot z^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial z}(z) = 1$:

$$\frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{\frac{x^2\sqrt{z}\frac{\partial}{\partial z}(z)}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1} = \frac{\partial}{\partial z}\left(y^2 z^3\right) + \frac{\frac{x^2\sqrt{z}1}{2\sqrt{-x^2z+1}} + \frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}$$

Apply the constant multiple rule $\frac{\partial}{\partial z}\left(c\cdot f\right)=c\cdot\frac{\partial}{\partial z}\left(f\right)$ with $c=y^2$ and $f=z^3$:

$$\frac{\partial}{\partial z}\left(y^2z^3\right)+\frac{\frac{x^2\sqrt{z}}{2\sqrt{-x^2z+1}}+\frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}=y^2\frac{\partial}{\partial z}\left(z^3\right)+\frac{\frac{x^2\sqrt{z}}{2\sqrt{-x^2z+1}}+\frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}$$

Apply the power rule $\frac{\partial}{\partial z}\left(z^n\right)=n\cdot z^{-1+n}$ with $n=3$:

$$y^2\frac{\partial}{\partial z}\left(z^3\right)+\frac{\frac{x^2\sqrt{z}}{2\sqrt{-x^2z+1}}+\frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}=y^2\left(3z^{-1+3}\right)+\frac{\frac{x^2\sqrt{z}}{2\sqrt{-x^2z+1}}+\frac{\sqrt{-x^2z+1}}{2\sqrt{z}}}{-x^2z+1}=3y^2z^2+\frac{1}{2\sqrt{z}\left(-x^2z+1\right)^{\frac{3}{2}}}$$

Thus, $\frac{\partial}{\partial z}\left(y^2z^3+\frac{\sqrt{z}}{\sqrt{-x^2z+1}}\right)=3y^2z^2+\frac{1}{2\sqrt{z}\left(-x^2z+1\right)^{\frac{3}{2}}}$

Therefore, $\frac{\partial^2}{\partial x\partial z}\left(xy^2z^3+\operatorname{asin}\left(x\sqrt{z}\right)\right)=3y^2z^2+\frac{1}{2\sqrt{z}\left(-x^2z+1\right)^{\frac{3}{2}}}$

Answer: $\frac{\partial^2}{\partial x\partial z}\left(xy^2z^3+\operatorname{asin}\left(x\sqrt{z}\right)\right)=3y^2z^2+\frac{1}{2\sqrt{z}\left(-x^2z+1\right)^{\frac{3}{2}}}$

See Here:
https://www.emathhelp.net/calculators/calculus-3/partial-derivative-calculator/?f=x*y%5E2*z%5E3%2B+asin+%28x+*+sqrt+%28z%29%29&var=x%2Cz%2Cy

$$\frac{\partial^2}{\partial x\partial z}\left(xy^2z^3+\operatorname{asin}\left(x\sqrt{z}\right)\right)=3y^2z^2+\frac{1}{2\sqrt{z}\left(-x^2z+1\right)^{\frac{3}{2}}}$$

NEXT, $\frac{\partial^3}{\partial x \partial z \partial y}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = \frac{\partial}{\partial y}\left(\frac{\partial^2}{\partial x \partial z}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right)\right)$

$$= \frac{\partial}{\partial y}\left(3y^2z^2 + \frac{1}{2\sqrt{z}(-x^2z+1)^{\frac{3}{2}}}\right)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial y}\left(3y^2z^2 + \frac{1}{2\sqrt{z}(-x^2z+1)^{\frac{3}{2}}}\right) = \left(\frac{\partial}{\partial y}\left(3y^2z^2\right) + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2z+1)^{\frac{3}{2}}}\right)\right)$$

Apply the constant multiple rule $\frac{\partial}{\partial y}(c \cdot f) = c \cdot \frac{\partial}{\partial y}(f)$ with $c = 3z^2$ and $f = y^2$:

$$\frac{\partial}{\partial y}\left(3y^2z^2\right) + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2z+1)^{\frac{3}{2}}}\right) = 3z^2\frac{\partial}{\partial y}\left(y^2\right) + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2z+1)^{\frac{3}{2}}}\right)$$

Apply the power rule $\frac{\partial}{\partial y}(y^n) = n \cdot y^{-1+n}$ with $n = 2$:

$$3z^2 \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2 z+1)^{\frac{3}{2}}}\right) = 3z^2\left(2y^{-1+2}\right) + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2 z+1)^{\frac{3}{2}}}\right) = 6yz^2 + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2 z+1)^{\frac{3}{2}}}\right)$$

The derivative of a constant is 0:

$$6yz^2 + \frac{\partial}{\partial y}\left(\frac{1}{2\sqrt{z}(-x^2 z+1)^{\frac{3}{2}}}\right) = 6yz^2 + (0)$$

Thus, $\frac{\partial}{\partial y}\left(3y^2z^2 + \frac{1}{2\sqrt{z}(-x^2 z+1)^{\frac{3}{2}}}\right) = 6yz^2$

Therefore, $\frac{\partial^3}{\partial x \partial z \partial y}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = 6yz^2$

Answer: $\frac{\partial^3}{\partial x \partial z \partial y}\left(xy^2z^3 + \operatorname{asin}\left(x\sqrt{z}\right)\right) = 6yz^2$

6(b):

We are given a point in cylindrical coordinates $(r, \theta, z) = \left(4, \frac{5\pi}{6}, 4\right)$ and asked to convert it to spherical coordinates (ρ, ϕ, θ).

Step 1: Recall the conversion formulas

From cylindrical (r, θ, z) to spherical (ρ, ϕ, θ):

$$\rho = \sqrt{r^2 + z^2}$$
$$\phi = \arccos\left(\frac{z}{\rho}\right)$$
$$\theta = \theta \quad \text{(same angle in both systems)}$$

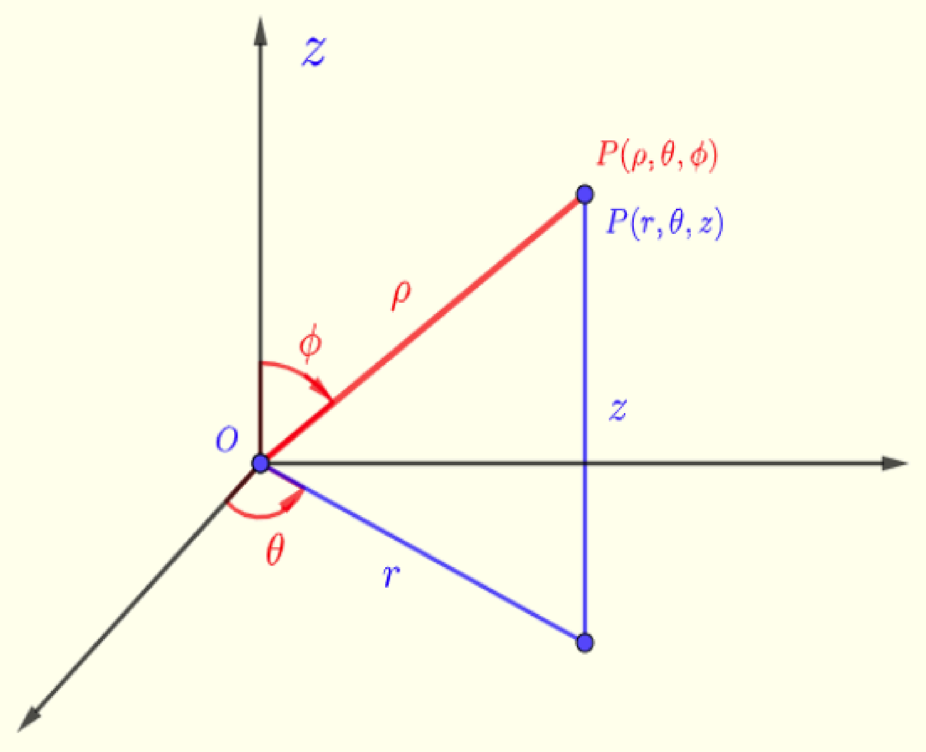

Fig.1 - Cylindrical and spherical coordinates

Step 2: Calculate ρ

$$\rho = \sqrt{r^2 + z^2} = \sqrt{4^2 + 4^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

Step 3: Calculate ϕ

$$\phi = \arccos\left(\frac{z}{\rho}\right) = \arccos\left(\frac{4}{4\sqrt{2}}\right) = \arccos\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

Step 4: θ remains the same

$$\theta = \frac{5\pi}{6}$$

Final answer:

The spherical coordinates are:

$$\boxed{\left(4\sqrt{2}, \frac{\pi}{4}, \frac{5\pi}{6}\right)}$$